



Koneru Lakshmaiah Education Foundation

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OPERATING SYSTEMS AND SYSTEMS PROGRAMMING (25CS2104E)

2026–27, Term-I

Project Problem Statement Submission Form

Section / Team Details

Section No.: 07

Team No.: 04

Project Title: **Linux Process Monitoring and Control System**

Team Members

Roll Number	Student Name	Signature
2520030401	Yashas	
2520030552	Gnaneshwar	

1. Abstract

The **Linux Process Monitoring and Control System** is a Linux-based systems programming project designed to monitor and manage running processes efficiently. The system provides important process information such as Process ID (PID), process name, process state, CPU usage, memory usage, and parent process details. It also allows users to control processes by sending signals such as SIGTERM and SIGKILL.

The project demonstrates key Operating Systems concepts including process management, process states, process identification, process control, signals, system calls, and the Linux /proc pseudo-filesystem. The application will be developed using the **C programming language** and Linux/POSIX APIs to retrieve process information and perform control operations.

The system provides a simple interface to view active processes, search for specific processes, monitor resource usage, and terminate processes when required. The expected outcome is a functional Linux utility that provides practical understanding of process management, system calls, and systems programming.

2. Problem Statement

Managing and monitoring multiple processes manually in a Linux environment can be difficult, especially when users need to identify resource-consuming or unresponsive processes. Existing commands such as ps, top, and kill provide process-management functionality, but beginners may find it difficult to understand how these utilities interact with the operating system.

The proposed project aims to develop a simple **Linux Process Monitoring and Control System** that displays important process information and provides process-control operations through a C-based application. The system will demonstrate how process information can be obtained and how signals can be used to control processes.

3. Objectives

1. To develop a Linux-based application for monitoring currently running processes and displaying important process information such as PID, process name, state, CPU usage, and memory usage.
2. To provide process-search and filtering functionality so that users can easily locate a particular process.
3. To implement process-control operations using Linux/POSIX mechanisms, including sending signals such as SIGTERM and SIGKILL to selected processes.
4. To demonstrate practical Operating Systems concepts such as process management, process states, system calls, signals, /proc filesystem, and user-kernel interaction.

4. Proposed Methodology

The project will be developed as a C-based Linux application. It will obtain process information from the Linux /proc filesystem and relevant POSIX/Linux APIs. The collected information will be processed and displayed through a simple command-line interface.

The major functionalities will include:

- Listing currently running processes.
- Displaying PID, process name, process state, CPU, and memory information.
- Searching for processes using PID or process name.
- Displaying detailed information about a selected process.
- Sending signals to terminate or control processes.
- Handling invalid PIDs and permission errors safely.

The application will use Linux system calls and POSIX interfaces for process-control operations. The project will be tested with different processes and scenarios to ensure that monitoring and control functions work correctly.

5. Operating Systems Concepts / Linux APIs Used

OS Concept / Linux API / System Call	Purpose in the Project
Process Management	Identify and manage currently running processes.
Process ID (PID)	Uniquely identify processes for monitoring and control.
/proc Filesystem	Obtain process information such as status, memory, and command details.
kill() System Call	Send signals to selected processes for process control.
Linux Signals	Control processes using signals such as SIGTERM and SIGKILL.
POSIX APIs	Provide standard interfaces for process and system-level operations.
Process States	Identify whether a process is running, sleeping, stopped, or terminated.

6. Individual Contribution

Roll Number	Student Name	Individual Responsibility
2520030401	Yashas	Process monitoring module, process information retrieval, /proc filesystem handling, and testing.
2520030552	Gnaneshwar	Process-control module, signal handling, user interface, integration, and documentation.

7. Tools / Platforms / Software Used

Tool / Platform / Software	Purpose
Linux / Ubuntu	Operating system and execution environment.
C / C++	Implementation of the process monitoring and control system.
GCC	Compilation of the C source code.
POSIX / Linux APIs	Process management and system-level operations.
/proc Filesystem	Retrieving runtime process information.
Git / GitHub	Version control and project repository management.

8. Expected Outcome

The project is expected to produce a functional Linux-based process monitoring and control application. The system will allow users to view active processes and obtain information such as PID, process name, process state, CPU usage, and memory usage.

The application will also provide process-search and process-control functionality, allowing users to send appropriate signals to selected processes. It will handle common errors such as invalid PIDs, permission restrictions, and processes that terminate during monitoring.

The project will demonstrate practical knowledge of Linux process management, process states, signals, system calls, /proc filesystem, and user-space interaction with the operating system. It will also provide hands-on experience in C-based systems programming.

9. GitHub Repository

GitHub Repository Link: <https://github.com/yashu-cyber/Linux-Process-Monitoring-and-Control-System>

Repository Created: _____

☐ Yes ☐ No

Faculty Name and Signature

Faculty Name: _____

Signature: _____

Remarks: _____

Faculty Use

Project Approved: _____

☐ Yes ☐ No

Date: _____

Faculty Signature: _____